

Osmolarity and osmotic pressure

First stage
February 2026

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Osmotic concentration (osmolarity)

Is the measure of solute concentration, defined as the number of osmoles (Osm) of solute per litre (L) of solution (osmol/L).

The osmolarity of a solution is usually expressed as **Osm/L** (pronounced "osmolar"), in the same way that the molarity of a solution is expressed as "M" (pronounced "molar"). Whereas molarity measures the number of moles of solute per unit volume litre(L) of solution. This value allows the measurement of the osmotic pressure of a solution, and the determination of how the solvent will diffuse across a semipermeable membrane (osmosis) separating two solutions of different osmotic concentration.

Unit

The unit of osmotic concentration is the **osmole**. This is unit of measurement that defines the number of moles of solute that contribute to the osmotic pressure of a solution. A **milliosmole (mOsm)** is 1/1,000 of an osmole. A **microsmole (μ Osm)** (also spelled **micro-osmole**) is 1/1,000,000 of an osmole.

Types of solutes

Osmolarity is distinct from molarity because it measures osmoles of solute particles rather than moles of solute. The distinction arises because some compounds can dissociate in solution, whereas others cannot.

Ionic compounds:

Ionic compounds such as salts, can dissociate in solution into their constituent ions, so there is not a one-to-one relationship between the molarity and the osmolarity of a solution. For example, sodium chloride (NaCl) dissociates into Na^+ and Cl^- ions. Thus, for every 1 mole of NaCl in solution, there are 2 osmoles of solute particles (1 mol/L NaCl solution is a 2 osmol/L NaCl solution). Both sodium and chloride ions affect the osmotic pressure of the solution.

Another example is magnesium chloride (MgCl_2), which dissociates into Mg^{2+} and 2Cl^- ions. For every 1 mole of MgCl_2 in the solution, there are 3 osmoles of solute particles.

Nonionic compounds do not dissociate, and form only 1 osmole of solute per 1 mole of solute. For example, a 1 mol/L solution of glucose is 1 osmol/L.

Calculations of osmolarity/osmotic pressure

Example I: Calculate osmolarity of Na_2HPO_4 solution

of $c = 21 \text{ g / l}$.

(AW of Na: 23, P: 31, O: 16, H: 1)

Calculations of osmolarity/osmotic pressure

Example II: 0.2M KCl solution is combined with an equal volume of 0.5 M glucose solution.

What is the resulting osmolarity?

After mixing:

Plasma osmolarity vs. osmolality

Plasma osmolarity can be calculated from plasma osmolality by the following equation:

Osmolality: defined as the number of osmoles (Osm) of solute per kilogram of solution (osmol/Kg).

$$\text{Osmolarity} = \text{osmolality} * (\rho_{\text{sol}})$$

where:

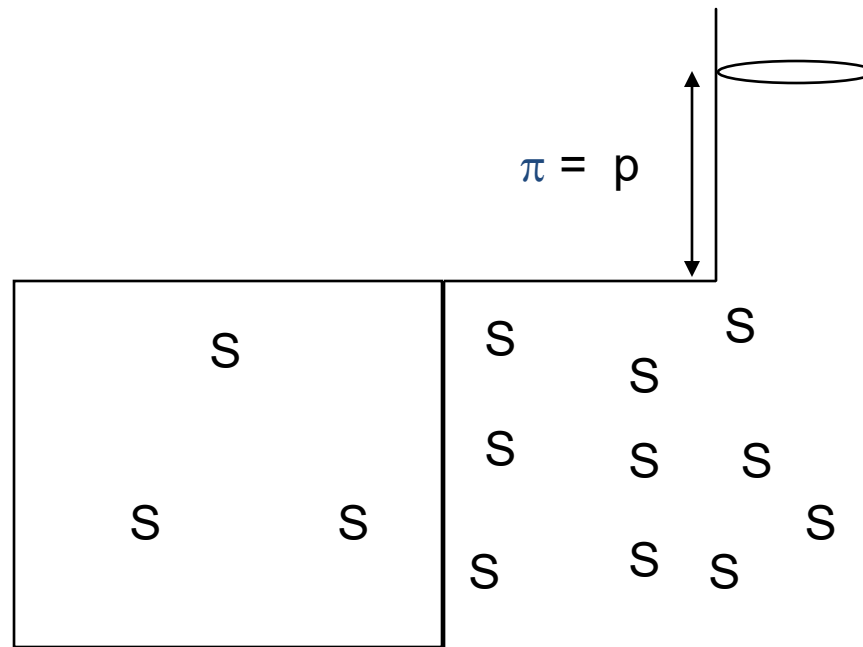
ρ_{sol} is the density of the solution in g/ml, which is 1.025 g/ml for blood plasma.

Net Osmotic Force Development

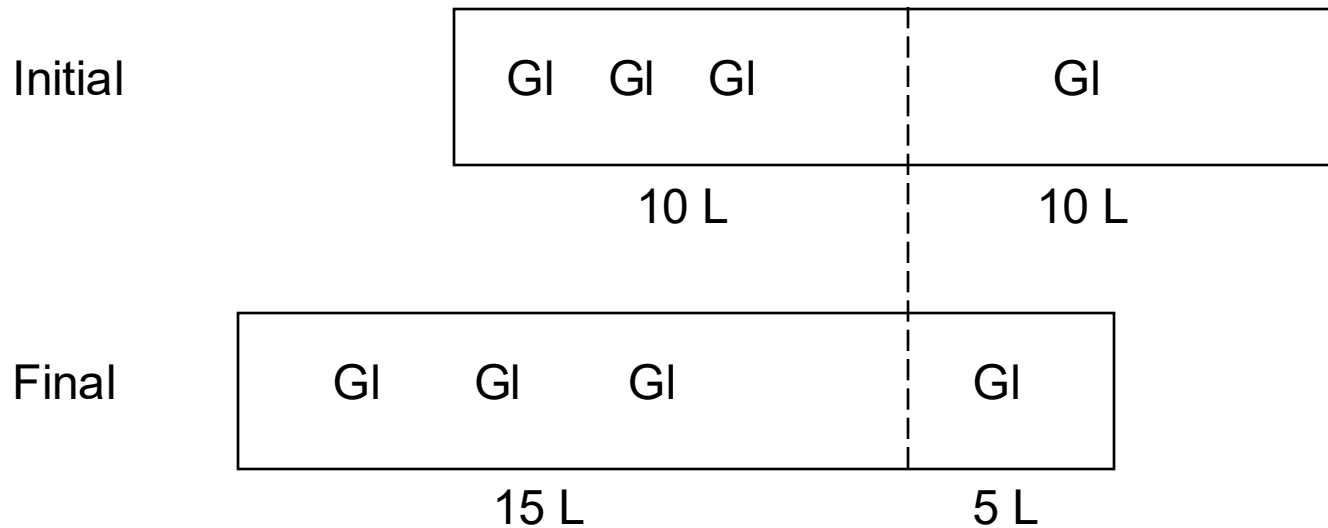
- Semipermeable membrane
- Movement some solute obstructed
- H_2O (solvent) crosses freely
- **End point:**
 - Water moves until solute concentration on both sides of the membrane is equal
 - **OR**, an opposing force prevents further movement

Osmotic Pressure (π)

- The force/area tending to cause water movement.



Glucose Example



Osmotic Concentration

- Proportional to the number of osmotic particles formed: $\text{Osm/L} = \text{moles} \times n$ (n , # of particles in solution)
e.g. 1 M NaCl = 2 M Glu in Osm/L
- Assuming complete dissociation:
 - 1mole of NaCl forms a 2 osmolar solution in 1L
 - 1mole of CaCl_2 forms a 3 osmolar solution in 1L

Milliequivalents, millimoles and milliosmoles

- **Example**
- Calculate the ideal osmolarity of 0.9 % NaCl injection?
- A distinction should be made between the terms osmolarity and osmolality
- **Osmolarity** is the milliosmoles of solute per liter of solution.
- **Osmolality** is the milliosmoles of solute per kilogram of solvent.
- Normal serum osmolality is considered within the range of 275 to 300 mOsmol/kg.
- Osmometers are commercially available for use in the laboratory to measure osmolality
- Abnormal blood osmolality can occur with shock, trauma, burns, electrolyte imbalance, hyperglycemia or renal failure

Clinical considerations of water and electrolyte balance

- Maintaining body water and electrolyte balance is an essential component of good health.
- Fluid and electrolyte therapy (oral or parenteral) is provided either for maintenance requirements or to replace serious losses or deficits.
- Example, a patient taking diuretics may simply require a daily oral potassium supplement along with adequate intake of water. Hospitalized patients commonly receive parenteral therapy of fluids and electrolytes to support ordinary metabolic functions.

In adult males, total body water ranges between 55% and 65% of body weight depending on the proportion of body fat. Adult women are about 10% less than adult men. Newborn infants have approximately 75% body water.

About 66.7 % ($\frac{2}{3}$) of adult's body water is intracellular and about 33.3 % ($\frac{1}{3}$) is extracellular

1500 ml of water per square meter may be used to calculate the daily requirements for adults.

On weight basis, 32 ml/kg for adults and 100-150 ml/kg for infants

Intracellular
fluid

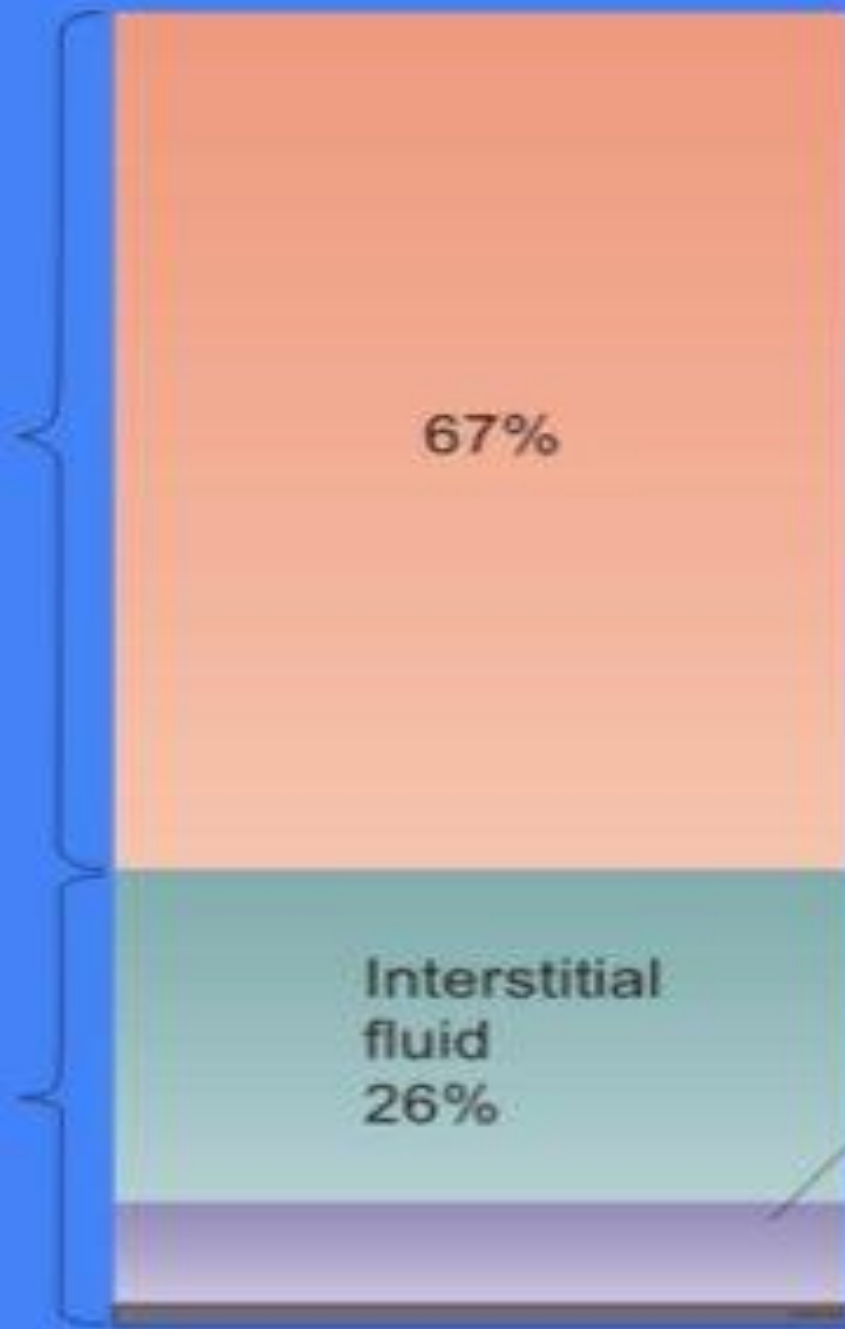
67%

Extracellular
fluid

Interstitial
fluid
26%

Intravascular fluid
(blood plasma) 7%

Cerebrospinal fluid
(less than 1%)



Clinical considerations of water and electrolyte balance

- Body fluids are extracellular (intravascular or interstitial) and intracellular (within cells)
- Osmolality of intracellular fluid and extracellular fluids is about equal (cell membrane) this value is about 290 mOsm/kg water.
- The plasma osmolality is approximated by the formula

$$plasma\ osmolality\ (mOsm / kg) = 2([Na] + [K])_{plasma} + \frac{[BUN]}{2.8} + \frac{[Glucose]}{18}$$

- Where Na and K are in mEq/L, and blood urea nitrogen (BUN) and glucose concentrations are in mg/100 ml (mg/dl)
- Example, Estimate the plasma osmolality from the following data: sodium 135 mEq/L; potassium 5 mEq/L; blood urea nitrogen, 14 mg/dL; and glucose, 90 mg/dL

If the patient has ingested ethanol, the ethanol level should be included in the calculated osmolality:

$$\text{Osmolality} = 2\{[\text{Na}^+] + [\text{K}^+]\} + [\text{Glucose}]/18 + [\text{BUN}]/2.8 + [\text{Ethanol}]/3.7$$

NOTE: Based on the molecular weight of ethanol the divisor should be 4.6 but empiric data shows that ethanol does not behave as an ideal osmole.

Diffusion:

Diffusion known as passive transport, is the process whereby substance moves from area of its higher concentration to region where it is less concentrated. During diffusion, external source of energy isn't required, for example;

- 1- the loss of perspiration from the body
- 2- the moisture flows from area of high concentration (the skin) to one of lower concentration (the air)

Osmosis:

Osmosis can be defined as the diffusion of water (solvent) through semi permeable membrane from weaker solution to stronger solution.

Example of osmosis can be seen when cucumber is placed in strong salt solution, the skin of the cucumber act as semi permeable membrane, the liquid inside the cucumber is quite dilute in comparison to that of the salt concentration. Therefore, osmosis takes place between solutions of two different concentrations, the flow of water, again from dilute to concentrated, is from the cucumber into the solution. Thus the cucumber shrinks and becomes pickle.

Homeostasis maintained by:

- Ion transport
- Water movement
- Kidney function

Tonicity

- Isotonic
- Hypertonic
- Hypotonic

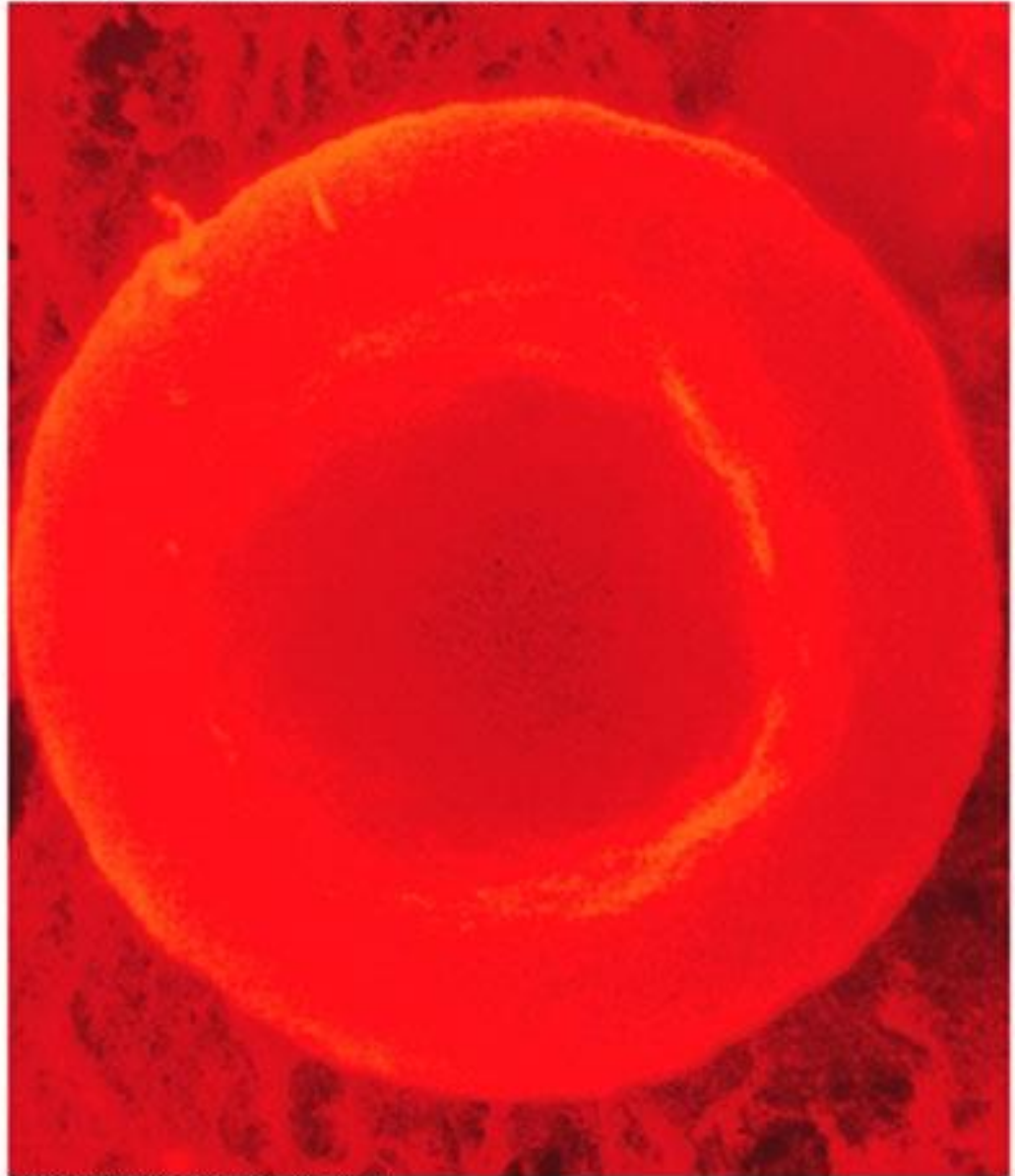
Isotonic solutions:

Are two solutions have the same solute concentration

The normal salt concentration of the blood is approximately equal to that of 0.9% NaCl solution. The common name for 0.9% NaCl solution is physiologic saline solution. The blood and physiologic saline solution are isotonic they have the same conc.

5.5% glucose solution is also approximately isotonic with body fluids.

Cell in Isotonic Solution



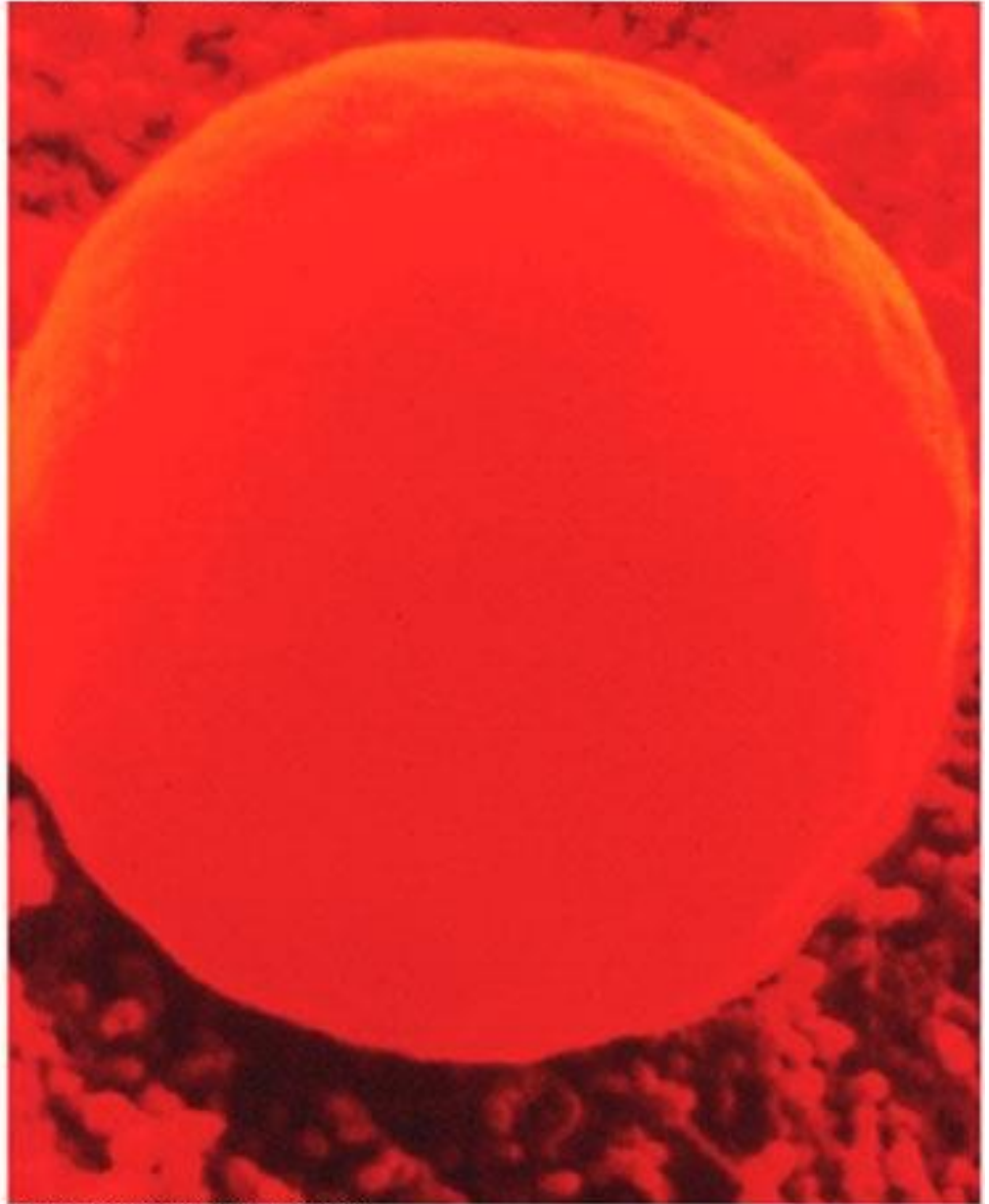
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Hypotonic solutions:

Hypotonic solution is one that contains lower solute concentration than that of another solution.

Distilled water and tap water are hypotonic compared with blood.

Cell in a hypotonic solution

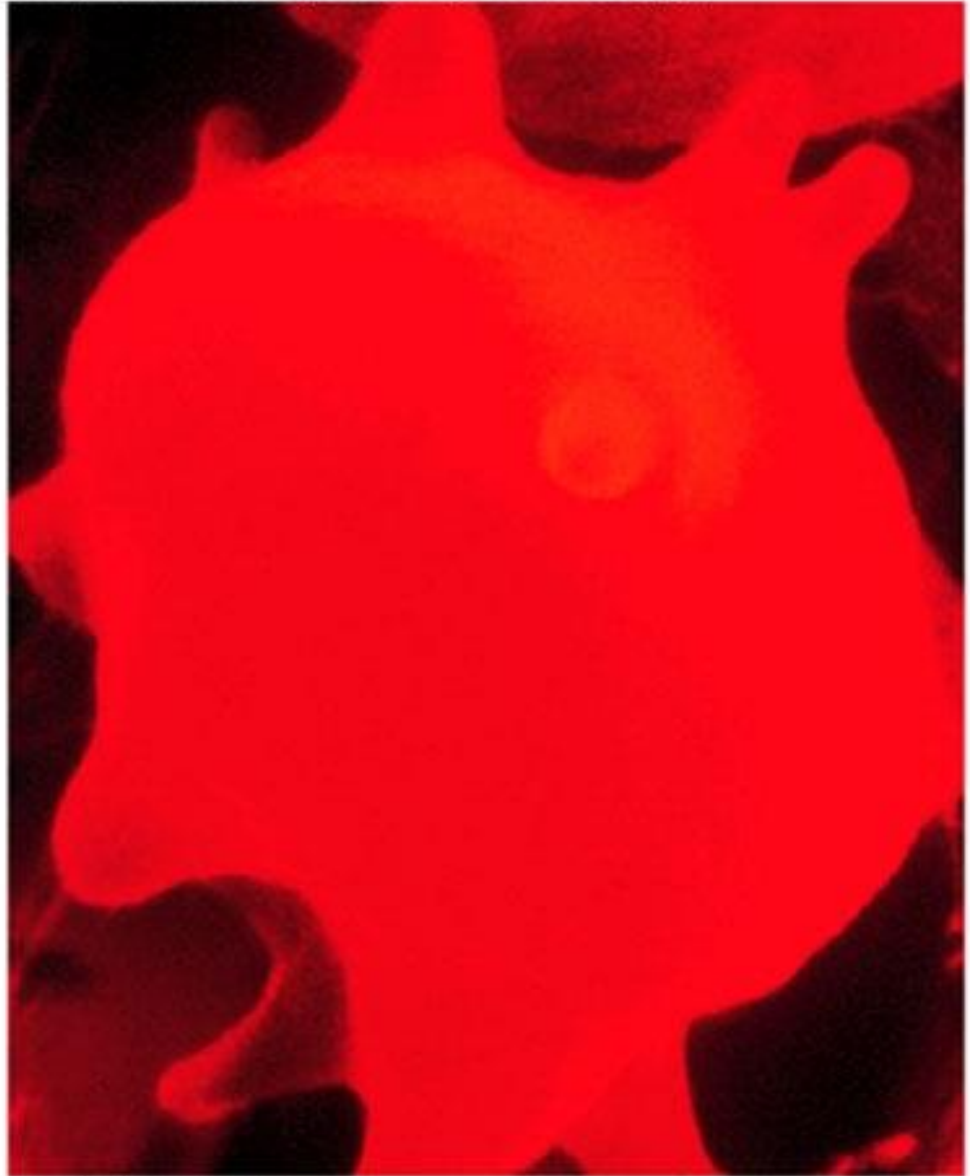


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Hypertonic solutions:

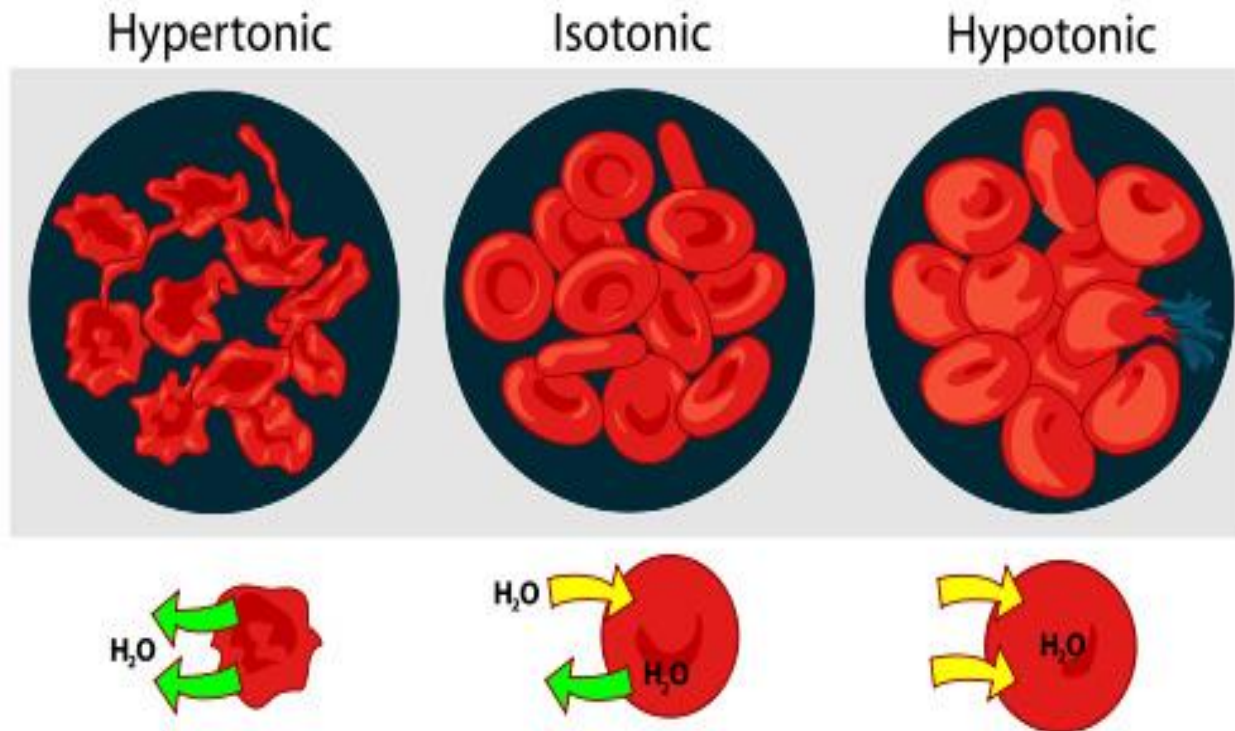
Hypertonic solution is one that contains higher solute concentration than the reference solution. 5% of NaCl solution or 10% glucose solution are examples of hypertonic solutions, when compared with blood.

Cell in a hypertonic solution



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Osmolarity is important in medicine



Isotonic: osmolarity = 0.3 osm/l (NaCl 0.15 mol/l)

- Hypotonic: osmolarity < 0.3 osm/l
- Hypertonic: osmolarity > 0.3 osm/l